

YFM FUSION COMPANION

Illustrated User Guide

Yu-Gi-Oh! Forbidden Memories - Windows x64 companion

Manual fusion tools work without an emulator

Optional read-only live and save integration for NTSC-U RetroArch/SwanStation

Independent fan-made utility - offline artwork with separate attribution

Install and start

Use the release ZIP - GitHub's source archive is for developers

Four steps

- Download a Windows x64 ZIP from the latest release on the Releases page.
- Extract the entire ZIP. Never run the program from inside the ZIP preview.
- Keep Resources, Documentation and Licenses beside the executable. Resources contains the database, artwork and research.
- Start YFM Fusion Companion.exe. The bottom line should report 722 cards and 25,146 resolved fusion pairs.

What is included

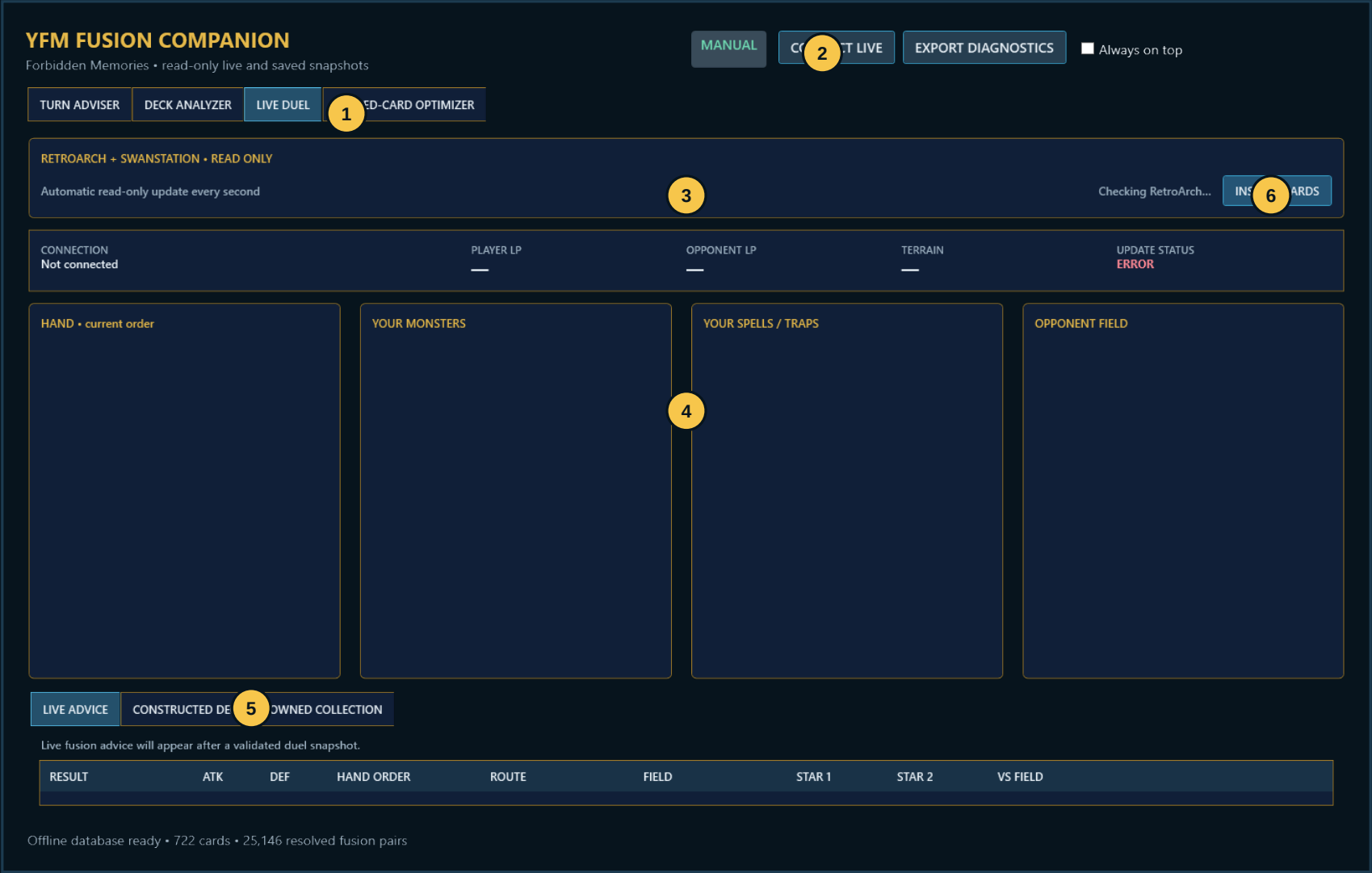
- The portable program is self-contained. Ordinary users do not install .NET.
- RetroArch/SwanStation is optional. It is only needed for Live Duel and automatic save discovery.
- Windows 10 or 11 x64 is required. The release is not code-signed, so SmartScreen may appear.
- Documentation includes the illustrated manual, searchable design guide and a source-code index. See CONTRIBUTING.md online to build from source.

Optional RetroArch setup

- RetroArch: Settings > Network > Network Commands = ON; Network Command Port = 55355.
- Leave Network RetroPad and stdin Commands OFF. Restart RetroArch once, then launch the NTSC-U game with SwanStation.
- The companion connects to 127.0.0.1 and sends read-only status and memory-read requests. Keep unsolicited inbound UDP 55355 blocked in Windows Firewall.

Interface map

The full window keeps source, display, troubleshooting, and workspace controls visible



- 1

Workspace tabs select the job.
- 2

Source badge and Compact Live are global.
- 3

Read-only connection and summary.
- 4

Hand and player/opponent field tables.
- 5

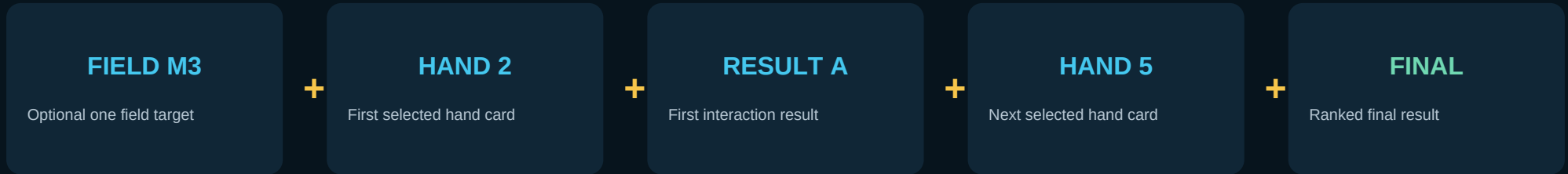
Advice, deck, and collection subtabs.
- 6

Inspector opens only when requested.

Turn Adviser

Enter only cards that actually exist in the current hand and player field

Legal selection order



Inputs and controls

- Hand: up to five ordered cards.
- Monster Field and Spell/Trap Field: up to five optional positions each.
- Analyze Turn ranks legal routes by final ATK, then DEF.
- Tab accepts the top autocomplete result and moves to the next input.
- Empty slots are ignored; duplicate copies remain distinct by slot.

Rules the route obeys

- The field card, when used, interacts with the first selected hand card.
- Only one occupied field position may participate.
- Intermediate results continue through later hand cards in the displayed order.
- Non-fusions that merely discard a material are omitted.
- Compatible equips are terminal; their bonus cannot survive a later monster fusion.
- Documented glitches can be included or excluded explicitly.

Deck Analyzer

Every physical five-card hand is examined for a complete 40-card deck

YFM FUSION COMPANION

Forbidden Memories • read-only live and saved snapshots

MANUAL

COMPACT LIVE

EXPORT DIAGNOSTICS

☐ Always on top

TURN ADVISER

DECK ANALYZER

LIVE DUEL

OWNED-CARD OPTIMIZER

DECK • up to 40 cards

0 / 40 cards • add cards below or load your saved deck

Add a card • name or number

Empty slot • optional

ADD COPY

REMOVE SELECTED COPY

ADVANCED • EDIT INDIVIDUAL DECK SLOTS

ANALYZE

LOAD CURRENT

CANCEL

CLEAR

Include documented glitch fusions

The analyzer reads the current validated save when you load the deck.

ANY FUSION / EQUIP

ATK ≥ 2500

ATK ≥ 2800

ATK ≥ 3000

EXPECTED BEST ATK

DEAD HAND

Fill any number of slots. A 40-card deck examines exactly 658,008 physical five-card hands.

RESULT

FINAL ATK

FINAL DEF

HAND CHANCE

HANDS

REPRESENTATIVE ROUTE

GLITCH

Offline database ready • 722 cards • 25,146 resolved fusion pairs

1

Enter up to 40 cards.

2

Load the validated saved deck in one click.

3

Run, cancel, clear, and glitch controls.

4

Exact probability and expected-ATK metrics.

5

Per-result hand chance and representative route.

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Live Duel

Validated state refreshes automatically every second

YFM FUSION COMPANION

Forbidden Memories • read-only live and saved snapshots

MANUAL

COMPACT LIVE

EXPORT DIAGNOSTICS

Always on top

TURN ADVISER

DECK ANALYZER

LIVE DUEL

OWNED-CARD OPTIMIZER

RETROARCH + SWANSTATION • READ ONLY

Automatic read-only update every second

Checking RetroArch... INS CARDS

CONNECTION
Not connected

PLAYER LP

OPPONENT LP

TERRAIN

UPDATE STATUS
ERROR

HAND • current order

YOUR MONSTERS

YOUR SPELLS / TRAPS

OPPONENT FIELD

LIVE ADVICE

CONSTRUCTED DECK

OWNED COLLECTION

Live fusion advice will appear after a validated duel snapshot.

RESULT

ATK

DEF

HAND ORDER

ROUTE

STAR 1

STAR 2

VS FIELD

Offline database ready • 722 cards • 25,146 resolved fusion pairs

- 1

Connection and read-only status.
- 2

Life Points, terrain, and update health.
- 3

Current ordered five-card hand.
- 4

Player and opponent active field positions.
- 5

Best legal live routes and both guardian-star chains; F#? preserves unknown opponent state.
- 6

Open the selected-card inspector.

Card Inspector

Select a live card, then inspect basic and advanced catalog data

YFM FUSION COMPANION

Forbidden Memories • read-only live and saved snapshots

MANUAL

COMPACT LIVE

EXPORT DIAGNOSTICS

☐ Always on top

TURN ADVISER

DECK ANALYZER

LIVE DUEL

OWNED-CARD OPTIMIZER

RETROARCH + SWANSTATION • READ ONLY

Automatic read-only update every second

CONNECTION
Not connected

PLAYER LP
—

OPPONENT LP
—

TERRAIN
—

HAND • current order

1

YOUR MONSTERS

YOUR SPELLS / TRAPS

LIVE ADVICE

CONSTRUCTED DECK

OWNED COLLECTION

Live fusion advice will appear after a validated duel snapshot.

RESULT

ATK

DEF

HAND ORDER

ROUTE

FIELD

STAR 1

STAR 2

V

Offline database ready • 722 cards • 25,146 resolved fusion pairs

CARD INSPECTOR

CLOSE

Select a card in the Live Duel tables to inspect its data.

2

3

4

- 1

Select a card in any live table.
- 2

Inspector slides over the right edge.
- 3

Basic type, ATK, and DEF appear first.
- 4

Advanced Data expands catalog relationships.

Recommended campaign decks

Six researched reference builds; opening-hand availability is not a duel win rate

DECKS & FREE DUEL DATABASE

Plan a 40-card build, find each missing card, or inspect every opponent reward table.

Search cards or duelists (3+ letters)

RECOMMENDED DECKS

FREE DUEL DATABASE

Six community-vetted strategies for the original game. Choose an icon to explore the full 40-card build.

MANUAL COLLECTION - isolated rendering fixture - Counts match required copies, including cards in your saved deck and chest.



Thunder

Campaign → endgame with power-ups

40 / 40 copies owned



Meteor Crossfire

Late campaign / final gauntlet

23 / 40 copies owned



Mercury Eclipse

Endgame alternative to Thunder

12 / 40 copies owned



Iron & Meteor

Farmed endgame / fewer materials

11 / 40 copies owned



Sand & Sorcery

Fan synergy → equip-heavy endgame

9 / 40 copies owned



Crimson Toolbox

Fan fusion web → upgraded endgame

14 / 40 copies owned

Thunder Tide • Campaign → endgame with power-ups

OPEN FREE DUEL DATABASE

Repeatable Thunder fusions, a single Umi, compatible power-ups and reliable removal

40 / 40 required copies owned • 0 missing copies

HOW TO PLAY AND UPGRADE

Start with available Dragon/Thunder materials. Replace weak three-material chains with stronger two-card recipes as you collect them. Umi + Megamorph + Dragon Treasure takes Twin-headed Thunder Dragon to 4,800 ATK over separate plays. Preserve Raigeki for threats your current monster cannot beat.

Crawling Dragon + Electric Snake → Twin-headed Thunder Dragon

Baby Dragon + Thunder Dragon → Twin-headed Thunder Dragon

A 2,800 ATK fusion alone is insufficient late in the game. The displayed setup odds require the field and equips to be present in the same opening hand; they do not assume an active Umi or predict a win.

CHECK ALL OPENING HANDS

CANCEL CHECK

Check the reference deck to see its opening-hand setup availability.

These metrics describe the complete reference list, including cards you may be missing. Setups may require several turns. Opponent fields, guardian stars, equips and responses

ADAPT THIS STRATEGY

COLLECTION

CLOSE

OFFLINE • READ ONLY

Uses owned cards as a starting deck; missing cards are replaced. Other strategies are still compared, so the best result can differ.

- 1

Distinct icons select each reference build.
- 2

Required copies already owned appear out of 40 beneath each icon.
- 3

Every missing card names its best Free Duel opponent, required rank group, and conditional drop chance.
- 4

Check all opening hands; cancel whenever needed.
- 5

Adapt a strategy to your collection; the optimizer compares other starts too.
- 6

Reference metrics can include missing cards; read sources and limits.

Free Duel database

Browse every Free Duel opponent and search cards or duelists after three characters

DECKS & FREE DUEL DATABASE

Plan a 40-card build, find each missing card, or inspect every opponent reward table.

Search cards or duelists (3+ letters)

RECOMMENDED DECKSFREE DUEL DATABASE

All 39 original Free Duel opponents - click a portrait for S/A POW, S/A TEC and B/C/D reward table. Each percentage is conditional on earning the named result-rank group. Weight divided by 2048 is the exact reward probability for that table.

 Simon Muran	 Teana	 Jono	 Villager1	 Villager2	 Villager3
 Seto	 Heishin	 Rex Raptor	 Weevil Underwood	 Mai Valentine	 Bandit Keith
 Shadi	 Yami Bakura	 Pegasus	 Isis	 Kaiba	 Mage Soldier
 Jono	 Teana	 Heishin	 Villager1	 Villager2	 Villager3

CLOSE

OFFLINE • READ ONLY

- 1

Search suggestions combine matching cards and duelists.
- 2

All 39 Free Duel opponents appear with their portraits and names.
- 3

Select a portrait without leaving the deck library.
- 4

Recommended Decks and Free Duel Database share one searchable screen.

Duelist reward tables

Opponent details open over the same window and preserve your place

DECKS & FREE DUEL DATABASE

Plan a 40-card build, find

RECOMMENDED DECK

All 39 original Free Duel 2048 is the exact reward

Simon Muran

Seto

Shadi

29 • Meadow Mage

Choose a result rank to inspect its complete reward table. Percentages are exact within that rank group.

S/A POW

S/A TEC

B/C/D

#	CARD	WEIGHT / 2048	DROP
009	Shadow Specter	150	7.32%
130	Weather Control	90	4.39%
422	Jinzo #7	88	4.30%
436	White Dolphin	88	4.30%
485	Korogashi	88	4.30%
516	Muka Muka	88	4.30%
563	Wretched Ghost of the Attic	88	4.30%
333	Sogen	60	2.93%
031	Koumori Dragon	46	2.25%
035	Dark Magician	46	2.25%
038	Gaia the Fierce Knight	46	2.25%
039	Curse of Dragon	46	2.25%
041	Celtic Guardian	46	2.25%
046	Griffore	46	2.25%
047	Torike	46	2.25%
048	Sangan	46	2.25%
059	Mammoth Graveyard	46	2.25%
065	Silver Fang	46	2.25%
074	Giant Soldier of Stone	46	2.25%
089	Catapult Turtle	46	2.25%
002	Mystical Elf	44	2.15%
006	Feral Imp	44	2.15%
010	Blackland Fire Dragon	44	2.15%

Villager3

Bandit Keith

Mage Soldier

CLOSE

CLOSE

OFFLINE - READ ONLY

- 1

The enlarged portrait and duelist name identify the selected opponent.
- 2

S/A POW, S/A TEC, and B/C/D tables remain separate.
- 3

Every row shows card number, name, weight, denominator, and exact conditional chance.
- 4

Select a card row to open its complete card and farming details.

Card and farming details

Card art, game data, password-shop data, and all drop sources appear together

DECKS & FREE DUEL DATABASE

Plan a 40-card build, find each missing card, or inspect every opponent reward table.

RECOMMENDED DECKS

FREE DUEL DATABASE

All 39 original cards are available in the 2048 is the

1



3

2

713 • Meteor B. Dragon

Dragon • Fire • Level 8 • ATK 3,500 / DEF 2,000

Guardian Stars •  Mars /  Sun

With the burning hide of a meteorite, this creature vaporizes all that it touches.

PASSWORD SHOP • unavailable

ALL FREE DUEL DROPS • HIGHEST CHANCE FIRST

DUELIST	RANK	DROP
Meadow Mage	S/A POW	0.98%
Jono 2nd	S/A POW	0.39%
Heishin	S/A POW	0.20%
Heishin 2nd	S/A POW	0.20%
Mountain Mage	S/A POW	0.10%

Each percentage is conditional on earning the named result-rank group. Weight divided by 2048 is the exact reward probability for that table.

CLOSE

1

The enlarged offline card image accompanies its in-game description.

2

Type, attribute, level, ATK, DEF, and both guardian stars are shown.

3

Password and starchip cost appear when the card is available in the shop.

4

Drop sources are ordered from highest to lowest conditional chance and link back to the duelist.

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Suggested deck and ordering

Save import is integrated into Deck Analyzer and Owned-Card Optimizer

YFM FUSION COMPANION

Forbidden Memories • read-only live and saved snapshots

Starting strategy: automatic • Compare all six builds: [YFM FUSION COMPANION](#)

exact analysis of 658,008 hands • 40/40 cards • seed 5850701 • ≥3500 setup available 0.0% • board clear drawn 0.0% • >4500 setup 0.0% • strong body 33.8% • no monster 0.0% • highest setup 3,000 ATK. Setups may require separate turns; not win probabilities. • support for Blue-eyes White Dragon: 0 useful field / 0 compatible equips • below the 1-field / 2-equip structure target.

OWNED COLLECTION

All types

Name

Add owned card

Empty slot • optional

ADD CARD

CUSTOM ART FOLDER...

CUSTOM SELECTED ART...

13 shown • 13 distinct • 39 copies

Baby Dragon

Dragon • ATK 1,200 / DEF 700 • ♀ / ♂

Owned 3 • manual quantity

Proposed deck: 3x

Black-lance Dragon

Dragon • ATK 1,500 / DEF 800 • ♀ / ♂

Owned 3 • manual quantity

Proposed deck: 3x

Blue-eyes White Dragon

Dragon • ATK 3,000 / DEF 2,500 • ♂ / ♀

Owned 3 • manual quantity

Proposed deck: 3x

Feral Imp

Fiend • ATK 1,300 / DEF 1,400 • ♂ / ♀

Owned 3 • manual quantity

Proposed deck: 3x

ATK ≥ 2800

0.00 %

ATK ≥ 2500

0.00 %

EXPECTED BEST

1,582

DEAD HAND

14.93 %

General campaign safety (33 opponents; final gauntlet excluded) • best found unless proof status explicitly says otherwise.

Modeled answers for 28/33 opponents • weakest modeled matchup 620 • opening answer coverage 83.94%. These are modeled safety scores, not duel-win probabilities.

Virtual purchase plan uses 9,005,806 of 0 Star Chips. Password redemption history is not stored by the game.

DECK

OUTCOMES

LIMITS

LOW VALUE

STAR CHIPS

THREATS

CARD ORDER

2 Alphabetical

3

COPIES	CARD	NO.	OWNED	BUY	TYPE	ATK	DEF	WHY INCLUDED
3	Baby Dragon	4	0	3	Dragon	1200	700	Monster / fusion material: Value depends on standalone strength and reachable fusion chains. 5 compatible deck partners; reaches B. Dragon Jungle King (2,100 ATK). Purchase 3 for 2,999,997 Star Chips.
3	Blackland Fire Dragon	10	0	3	Dragon	1500	800	Monster / fusion material: Value depends on standalone strength and reachable fusion chains. 4 compatible deck partners; reaches B. Dragon Jungle King (2,100 ATK). Purchase 3 for 240 Star Chips.
3	Blue-eyes White Dragon	1	0	3	Dragon	3000	2500	Monster / fusion material: Value depends on standalone strength and reachable fusion chains. 0 compatible deck partners; base ATK 3,000. Purchase 3 for 2,999,997 Star Chips.
3	Feral Imp	6	0	3	Fiend	1300	1400	Monster / fusion material: Value depends on standalone strength and reachable fusion chains. 2 compatible deck partners; reaches Koumori Dragon (1,500 ATK). Purchase 3 for 420 Star Chips.
3	Hitotsu-me Giant	3	0	3	Beast-Warrior	1200	1000	Monster / fusion material: Value depends on standalone strength and reachable fusion chains. 2 compatible deck partners; reaches Koumori Dragon (1,500 ATK). Purchase 3 for 310 Star Chips.

Offline database ready • 722 cards • 25,146 resolved fusion pairs

1

Enter or load owned quantities; card artwork works offline.

2

Alphabetical is the default card order.

3

Choose card number, ATK or DEF; the choice is remembered.

4

ATK and DEF sort highest first; card numbers lowest first.

5

Campaign summary keeps uncertainty visible.

6

Fields, equips and removal are scored as coherent support; setups may need separate turns.

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Compact Live

A narrow right-side companion that leaves the central duel area visible



- 1

Result name and effective ATK.
- 2

Hand routes use plain slot numbers.
- 3

F(1)-F(5) are monster positions; F(6)-F(10) are spell/trap positions.
- 4

20px stars sit beneath Result; warm/red glows clarify choices and known enemy relations.
- 5

PIN stays above the game; FULL restores the normal window.

Compatibility, privacy, and troubleshooting

Compatibility boundary

- Manual adviser, analyzer, and optimizer work from the bundled offline catalog.
- Live decoding and automatic discovery are validated for NTSC-U / SLUS-01411 in RetroArch/SwanStation.
- Other regions, revisions, emulators, cores, and modified games are not claimed compatible; use manual tools.
- RetroAchievements Hardcore Mode works because the companion never uses save states.

Read-only and private by design

- No game input, memory writes, save edits, config edits, telemetry, updater, ads, or online API.
- Local settings store window bounds, compact/topmost choices, and the last selected save path.
- Diagnostics are created only on request and exclude gameplay/card values.
- No ROM, BIOS, emulator or memory-card save is distributed. Bundled artwork has separate attribution.

When something is unavailable

- ERROR or DISCONNECTED: confirm Network Commands, port 55355, firewall scope, SwanStation, and that RetroArch was restarted.
- NOT IN DUEL: connectivity is healthy; enter a duel. Stale duel structures are intentionally hidden in story mode.
- Save looks old: save in-game, close content so the emulator flushes the file, then Refresh in the analyzer or optimizer.
- Wrong region/core: Live Duel may be rejected safely, but every manual feature remains available.
- For support, use Export Diagnostics; review the text yourself before sharing it.